

Calibrated date ranges for the Pentateuchal sources, and a direct measurement of what archaizing can buy: Biblical Hebrew and Ancient Greek

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Abstract

Linguistic dating of ancient texts rests on an assumption that has never been measured: that a late author cannot convincingly imitate an earlier stage of the language. If imitation is cheap, every early date is worthless. This paper measures the price of imitation, and uses a model calibrated against that price to place the Pentateuchal sources in time. Features are extracted from 146,059 words of independently anchored Biblical Hebrew — 25 prophetic and post-exilic books whose dates rest on synchronisms with Assyrian, Babylonian and Persian records — partitioned into 285 passages of about 500 words so that high-frequency morphosyntax can be estimated within each passage rather than only across whole books. A date model over 579 features, weighted to counteract the corpus's post-exilic skew and variance-matched so that its output spans the range of real dates, achieves a leave-one-book-out mean absolute error of 114 yr against 137 yr for the constant predictor, a rank correlation of +0.70 with true date, and correct chronological ordering of 74.9% of 295 book pairs; a permutation null that re-runs the entire pipeline on shuffled dates gives $p = 0.005$. Applied to the documentary sources with distribution-free conformal intervals, the model places the Priestly, Deuteronomic and JE composites after the exile with probability ≥ 0.84 , and recovers the classical relative sequence $JE \rightarrow D \rightarrow P$ without having been trained on any of them. The archaizing question is then measured rather than assumed away, in two experiments that give sharply different answers. Replacing every Late Biblical Hebrew form in six securely dated late books with its Classical counterpart — total lexical archaizing, beyond what any scribe sustains — moves their estimated dates by a mean of only +11 yr, because no single lexical item is worth more than 12.2 yr per standard deviation and the model's remaining leverage sits in morphology and syntax. But in Ancient Greek, where archaizing was practised rather than simulated, the Second Sophistic Atticizers — professional imitators of Classical Attic trained in a formal curriculum of Attic prose composition — are displaced by a mean of -216 yr under a model trained only on non-archaizing Greek, with 13 of 14 placed earlier than they were written. The price of archaizing therefore depends entirely on its mechanism: substituting archaic vocabulary is cheap and nearly undetectable in its effect on date, while reconstructing an archaic grammar is expensive to perform and can move an estimate by centuries. This reframes the standing objection to linguistic dating. An archaizing explanation for an apparently early Hebrew text requires sustained morphosyntactic imitation of a kind for which the Second Temple period offers no attested institutional support, not the lexical substitution the Classical/Late Biblical Hebrew literature describes. Code and data are publicly available.

Introduction

Scholars in biblical studies, classical philology, and Near Eastern linguistics routinely use internal linguistic evidence to constrain the dates of ancient texts. The transition from Classical Biblical Hebrew (CBH; also called Standard Biblical Hebrew) to Late Biblical Hebrew (LBH) is among the best-documented diachronic shifts in any ancient language [1–3], and comparable transitions are well attested in Ancient Greek [4] and Akkadian. Despite this rich empirical tradition, linguistic dating almost never produces calibrated probability estimates; scholars identify features as “archaic” or “late” and assign texts to qualitative categories, but the resulting claims carry no quantified uncertainty.

Underneath the practice sits an assumption that is stated often and tested never. The whole enterprise presupposes that a late author cannot convincingly write like an early one. If deliberate archaizing were easy — if a Persian-period scribe could adopt Classical vocabulary and thereby acquire a Classical date — then every linguistic argument for the antiquity of a biblical text would be worthless, and the objection would apply with equal force to computational and traditional methods. The dialectal school has pressed exactly this point for two decades [5, 6], and the mainstream reply has been that sustained archaizing is difficult. Both positions are assertions. Neither side has put a number on it.

This paper puts a number on it, in two independent ways, and then uses a model whose archaizing sensitivity is known to place the Pentateuchal sources in time. The two contributions are of equal weight, and the second is what makes the first worth believing.

Contribution 1: calibrated date ranges. We build a date model on 146,059 words of independently anchored Hebrew, partitioned into 285 passages of about 500 words. The partition matters more than it sounds: at book level the corpus supplies 25 observations for hundreds of candidate features, a regime in which nothing can be estimated honestly. At passage level the same corpus supplies 285 observations, and the features that survive are the high-frequency morphological and syntactic rates that can actually be measured in a few hundred words. The model reports distribution-free conformal intervals rather than point dates, and its intervals are calibrated on out-of-sample residuals rather than assumed.

Contribution 2: the price of archaizing. We then measure what the model can be fooled by, twice, by two routes that turn out to disagree informatively. Because we can perform archaizing ourselves, the ground truth is known exactly: we take securely dated *late* books, replace their LBH forms with CBH counterparts at controlled rates, re-extract every feature from the altered text, and re-date it with a model that never saw it. This prices the *lexical* mechanism, and the price is low. Separately, Ancient Greek supplies what Hebrew cannot — a set of texts whose authors really did archaize, deliberately and expertly, with securely known composition dates. The Second Sophistic Atticizers wrote imitation Classical Attic four to seven centuries after the Classical period, taught by a formal curriculum, and their imitation was good enough to be read as a stylistic model for a millennium. This prices the *morphosyntactic* mechanism, and the price is an order of magnitude higher.

That contrast is the paper’s answer to the standing objection. Archaizing is not uniformly cheap or uniformly hard; its cost depends on how deep into the grammar the imitation goes, and the two depths are separated by roughly a factor of twenty in their effect on an estimated date. Whether a given text could have been archaized therefore turns on what its author was trained to do, which is a historical question about scribal education rather than a statistical one.

The results are bounded in both directions, and we state the bounds plainly. The model orders books in time far better than chance and places the Pentateuchal sources after the exile with high probability, but its 138 yr 68% intervals are wide, and it should not be read as resolving dates within the Persian period. What it does resolve is the pre-exilic versus post-exilic question, which is the question the Pentateuchal debate has actually turned on.

The paper is organized as follows. The Background section reviews the CBH/LBH distinction, the dialectal challenge, and the circularity problem. The Data section describes the anchored corpus, the passage-level design, and feature extraction. The Model section presents the weighted ridge estimator, the variance-matching step, and conformal intervals. The Validation section documents a failure mode that inflates reported accuracy in this literature, then reports leave-one-book-out performance against a permutation null. The Results section dates the undated units. The Archaizing section reports both measurements. The Discussion addresses limitations and scholarly context.

Background: diachronic Biblical Hebrew

The CBH/LBH distinction

Classical Biblical Hebrew (CBH) is the language of the pre-exilic prophetic corpus and early narrative material (8th–7th centuries BCE): Amos, Hosea, early Isaiah, and the narrative strands of Samuel–Kings. Late Biblical Hebrew (LBH) characterizes the post-exilic prose of Ezra, Nehemiah, Chronicles, Esther, and Daniel (5th–2nd centuries BCE).

The transition is marked by well-studied lexical, morphological, and syntactic changes [2,3,7,8]. At the lexical level, the first-person singular pronoun shifts from *ankī* to *anī*; the relative pronoun *ašer* gives way to the enclitic *še-*; the negator *lō'* is supplemented by and partly replaced by *'ēn*. At the morphological level, the wayyiqtol declines in frequency while the *hāyāh* + participle periphrastic increases. At the syntactic level, infinitive-absolute constructions rise, and sentence structure shifts toward analytic patterns associated with Aramaic influence [5,6].

The consensus view [1,9–11] holds that these features shift monotonically along the temporal axis, constituting a reliable if noisy diachronic signal. The present study adopts this view as a working assumption and then tests the principal alternative to it directly.

The diachronic versus dialectal debate

Young, Rezetko, and Ehrensverd [5] argue that the CBH/LBH contrast may reflect geographic dialect, scribal tradition, or literary register rather than chronology. The Qumran sectarian texts (~150 BCE) deliberately employ CBH vocabulary and morphology while simultaneously using Aramaic loan-words — a pattern they interpret as literary register rather than temporal reality. Hurvitz and the diachronic mainstream [7] counter that the dialectal alternatives require additional auxiliary hypotheses, and that the Qumran archaism case presupposes the original CBH features had already receded, confirming rather than undermining the diachronic interpretation.

This debate has been conducted almost entirely through argument. Its empirical core is a quantitative question — how much apparent antiquity does deliberate archaizing actually purchase? — and that question is answerable. A model that assigns dates can be fed texts whose archaizing status is known, either because the archaizing was performed by the investigator or because the author's imitative program is a matter of literary-historical record. The Archaizing section reports both experiments. We regard this as the substantive contribution of the paper to the dialectal debate: not a resolution of it, but a measurement that either side can now argue from.

The circularity problem

Dating methods that use features identified by comparing texts of assumed date risk circularity: the assumed dates shape the feature set, and the feature set is then used to infer unknown dates. Three safeguards are adopted.

First, the training corpus is restricted to books with *independent* historical anchoring via synchronisms with datable rulers in Assyrian, Babylonian, and Persian king-lists. Torah texts are excluded from training entirely, so all Torah date estimates are genuine out-of-sample predictions.

Second, no feature is selected using date information. The feature set is fixed in advance by frequency rank in the corpus as a whole (Data section), not by correlation with date. Feature-by-date screening was evaluated and rejected: at this dimensionality the false discovery proportion among screened features is high enough that screening manufactures apparent signal (S2 Table).

Third, all reported performance comes from leave-one-*book*-out cross-validation in which standardisation, weighting, regularisation-parameter selection and interval calibration are performed inside each fold, and significance is established by a permutation null that re-runs the entire pipeline on shuffled dates.

Why an anchored corpus. The Hebrew prophetic books offer a form of historical anchoring available nowhere else in the Hebrew Bible: their superscriptions and internal chronological notices can be cross-checked against records entirely external to the biblical tradition. Ezekiel explicitly dates thirteen of his visions to regnal years of the Babylonian exile — “the fifth year of King Jehoiachin’s captivity” (Ezek 1:2),

“the sixth year” (8:1), and so forth — which correlate with Babylonian administrative texts independently dating Jehoiachin’s deportation to 597 BCE. Other eighth- and seventh-century prophets (Amos, Hosea, Isaiah, Micah, Zephaniah, Nahum) are anchored by superscriptions naming Judahite and Israelite kings whose reigns are documented in the Assyrian and Neo-Babylonian annals. No comparable external anchoring exists for the Torah or the poetic books: their dates derive entirely from literary-critical argument, which is precisely the circularity this study seeks to avoid.

Asymmetric error and conservative upper bounds on antiquity. If the accepted dates for the anchored training texts were wrong, the error would be asymmetric: these texts can be *no older* than the historical events they reference, but they could in principle be *younger*. A training corpus misdated in the direction of being too early would calibrate the feature-to-date mapping at too early a point on the temporal axis, so unknown texts dated against that calibration would also be estimated as too early. Any systematic error in the training dates therefore propagates in a single direction: all model-derived dates shift toward greater recency. The dates reported here should be read as *conservative upper bounds on antiquity*. A text dated to 400 BCE may be more recent; the linguistic evidence provides no support for a date earlier than the estimate.

Prior computational work: attribution versus dating

Computational approaches to textual attribution and stylometric dating have a substantial history, from Mosteller and Wallace’s Bayesian analysis of the Federalist Papers [12] through the stylometric tradition surveyed by Holmes [13] and the function-word methods of Burrows [14]. Among work applied specifically to the Hebrew Bible, Faigenbaum-Golovin et al. [15] provide the most methodologically rigorous prior computational treatment. Their Higher Criticism discrepancy method tests whether the distribution of lemma n-gram frequencies in a target text deviates from the null that it was produced by the same process as a reference corpus. Applied to 50 ground-truth chapters spanning D, the Deuteronomistic History, and P, it achieves 84% attribution accuracy in leave-one-out validation and yields interpretable discriminating lemma lists.

The present framework answers a different question, and three differences follow from that.

Continuous dates versus categorical assignment. The Higher Criticism method assigns each text to the most likely of three corpora. Because D and the Deuteronomistic History are both dated to the late monarchic or exilic period, two texts composed at 650 and 590 BCE receive identical labels; no intra-school temporal resolution is possible. The model used here returns a date with a calibrated interval, recovering whatever temporal resolution the data support and quantifying the rest as uncertainty.

Period features versus topic features. Faigenbaum-Golovin et al. operate on lemma and lemma n-gram frequencies, a representation in which content words — *gold* in Tabernacle chapters, *king* in regnal narratives — dominate. Content-word frequency is sensitive to topic and genre as much as to date. The feature set here is deliberately dominated by high-frequency grammatical rates (verbal aspect and stem distributions, part-of-speech transition rates, clause-relation profiles) that vary with period rather than subject matter. This is what permits comparison of poetry against narrative, which a bag-of-words representation cannot do.

A measured archaizing sensitivity. Attribution methods offer no mechanism for distinguishing a genuinely ancient text from one composed in a deliberately archaic register: a Persian-period author who archaizes to D norms is assigned to D regardless of composition date. Rather than claim immunity, we measure the sensitivity directly and report it as a quantity (Archaizing section). A reader who believes archaizing was widespread can take the measured exchange rate and compute for themselves how much of any reported date it could account for.

These are differences of purpose rather than deficiencies in either approach. The ideal analysis of a disputed text would apply both: attribution output can set register priors, and date estimates constrain the plausible compositional settings for attribution decisions.

Data: corpus and feature extraction

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The BHSa database

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All Hebrew analysis uses the Biblia Hebraica Stuttgartensia Amstelodamensis (BHSa) database [16], accessed through Text-Fabric [17]. BHSa supplies a complete morphological and syntactic annotation of the Masoretic text: for every word, a lexeme identifier, part of speech, verbal tense/aspect and stem, person, number, gender, pronominal suffix and state; and for every phrase and clause, a type, a syntactic function, and a relation to its governing clause. These annotations are the product of decades of work by the Eep Talstra Centre for Bible and Computer and are independent of any dating hypothesis.

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Training corpus

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The training corpus comprises 25 units spanning 760–167 BCE, listed with their anchors in Table 1. Every date rests on a synchronism external to the biblical tradition or on an event whose date is fixed by external record. Books with disputed internal composition histories are split into the conventional scholarly divisions (Isaiah into three, Zechariah into two) and the oracular core of Jeremiah is separated from its Deuteronomistic prose frame, which is treated as an undated target rather than an anchor.

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Table 1. The anchored training corpus. Dates are BCE. Every date rests on a synchronism or event external to the biblical tradition. “Pass.” is the number of ~500-word passages.

Unit	Date	Words	Pass.	External anchor
Amos	760	2,780	5	Superscription: Uzziah of Judah / Jeroboam II of Israel
Hosea	745	3,146	6	Superscription: Uzziah to Hezekiah / Jeroboam II
Micah	725	1,895	4	Superscription: Jotham, Ahaz, Hezekiah
Isaiah_1	720	13,504	26	Superscription: Uzziah to Hezekiah; Assyrian campaigns
Nahum	650	746	1	Fall of Nineveh, 612 BCE
Zephaniah	630	1,037	2	Superscription: Josiah
Habakkuk	605	897	2	Neo-Babylonian ascendancy
Jer_oracle	605	14,539	29	Superscription: Josiah to Zedekiah; 597 and 586 BCE
Obadiah	585	392	1	Edomite conduct at the fall of Jerusalem, 586 BCE
Ezekiel	580	26,182	51	Thirteen regnal dates from Jehoiachin’s deportation, 597 BCE
Lamentations	570	1,945	4	Destruction of Jerusalem, 586 BCE
Isaiah_2	545	5,738	11	Cyrus named; fall of Babylon, 539 BCE
Haggai	520	877	2	Regnal dates, second year of Darius I, 520 BCE
Zechariah_1	518	2,484	5	Regnal dates, second to fourth year of Darius I
Malachi	460	1,187	2	Persian governor (<i>peha</i>); second-temple cult
Joel	400	1,318	3	Second-temple setting; post-exilic references
Jonah	400	985	2	Late linguistic profile; post-exilic novella
Isaiah_3	400	3,689	7	Restored temple cult; Persian period
Ezra	380	5,268	10	Persian regnal dates, Cyrus to Artaxerxes
Nehemiah	380	7,842	15	Twentieth year of Artaxerxes I, 445 BCE
Zechariah_2	350	1,987	4	Greek (<i>Yawan</i>) reference; late Persian
Chronicles	330	35,330	69	Genealogies to the Persian period; Cyrus decree
Esther	300	4,621	9	Achaemenid court setting; Xerxes
Ecclesiastes	250	4,233	8	Persian loanwords (<i>pardes</i> , <i>pitgam</i>)
Daniel	167	3,437	7	Antiochene persecution, 167 BCE

Passage-level design

The single most consequential design choice in this paper is that features are extracted from passages of about 500 words rather than from whole books.

At book level the corpus offers 25 observations. Any feature set large enough to describe Hebrew morphosyntax then has more dimensions than there are data points, and the standard responses — screening features by their correlation with date, or reducing dimension using the date labels — both use the labels before cross-validation and inflate apparent performance. Worse, a whole-book rate is a single number summarising tens of thousands of words, so the model cannot distinguish a book whose passages are internally consistent from one that is a composite.

Partitioning at ~ 500 words yields 285 passages from the same 146,059 words. The gain is not a free multiplication of sample size: passages from one book are correlated, and the appropriate measure is the effective sample size $n_{\text{eff}} = n / (1 + (\bar{m} - 1)\rho_I)$ where ρ_I is the intraclass correlation and $\bar{m}$ the mean passages per book. Across chunk sizes from 200 to 1000 words the median intraclass correlation runs 0.20–0.32 and the resulting effective sample size is three to seven times the book-level figure (S2 Table). That is the honest gain, and it is what makes a feature set of this width estimable at all.

Passage size trades two effects against each other. Smaller passages give more observations but sparser features: a lexical item occurring once per thousand words is absent from most 200-word passages, and its rate is then not a measurement but a coin flip. Larger passages give denser features but fewer observations. We searched passage sizes of 200, 300, 400, 500, 700 and 1000 words crossed with three feature sets and with calibration on and off, 36 configurations in all, selecting on out-of-sample rank correlation under leave-one-book-out (S3 Table). The selected configuration uses ~ 500 -word passages; performance is flat between 400 and 1000, so the choice is not delicate.

Passages respect verse boundaries: words are accumulated verse by verse until the target is reached. A trailing remainder shorter than half the target is merged into the preceding passage rather than kept as a short, high-variance observation.

Feature extraction

The feature set is fixed by frequency rank computed over the whole BHSa corpus, without reference to date. It comprises 579 features in six families:

1. **Lexical rates** (250 features): occurrences per 1000 words of each of the 250 most frequent lexemes in the corpus. Because selection is by corpus frequency, these are dominated by function words — articles, prepositions, conjunctions, pronouns — rather than by the rare diagnostic vocabulary of the CBH/LBH literature. This is deliberate; the sparsity constraint above rules out the rare items, and the Archaizing section shows the consequence.
2. **Part-of-speech transition rates** (207 features): rates of each ordered pair of parts of speech across adjacent words, capturing local syntactic profile without reference to any particular construction.
3. **Verb morphology** (57 features): distributions over tense/aspect (qatal, yiqtol, wayyiqtol, participle, infinitive, imperative), stem (qal, nifal, piel, hifil, hitpaal, pual, hofal), and their interaction.
4. **Phrase and clause structure** (40 features): phrase type and syntactic function rates, and clause-relation rates from the BHSa clause hierarchy.
5. **Agreement and suffix morphology** (20 features): person, number, gender and pronominal-suffix rates.
6. **Structural** (5 features): mean phrase and clause length, clauses per passage, and type-token ratio.

A character and part-of-speech n-gram extension adding 689 further features was built and evaluated (S3 Table). It does not improve out-of-sample performance at the selected passage size and is not used in the reported model; it is retained in the repository for replication.

Features whose standard deviation is zero on the training fold, or which are missing in more than 20% of passages, are dropped inside the fold. Remaining missing values are imputed at the training-fold median.

Model

Weighted ridge regression

Let $\mathbf{X}$ be the $n \times p$ matrix of passage features and $\mathbf{y}$ the vector of book dates, each passage inheriting its book's date. Features are centred and scaled on the training fold. The estimator is ridge regression,

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \sum_{i=1}^n w_i (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \lambda \|\boldsymbol{\beta}\|^2, \quad (1)$$

solved in dual form, $\hat{\boldsymbol{\beta}} = \mathbf{A}^\top (\mathbf{A}\mathbf{A}^\top + \lambda \mathbf{I})^{-1} \tilde{\mathbf{y}}$, where $\mathbf{A}$ is the weighted standardised design. Because $p > n$ on the Hebrew side, the dual form is the cheaper route, and one eigendecomposition of $\mathbf{A}\mathbf{A}^\top$ serves the entire λ sweep. λ is selected by an inner leave-one-book-out loop nested inside each outer fold, never using the held-out book.

Inverse-density weighting

The anchored corpus is not uniform in time. Post-exilic material outnumbers pre-exilic material, and an unweighted fit is pulled toward the dense region: in the unweighted model no book receives a confidently pre-exilic estimate, which is a property of the sampling rather than of Hebrew. Weights are therefore taken inversely proportional to a kernel estimate of the date density,

$$w_i \propto \left[\sum_j \exp \left(-\frac{1}{2} \left(\frac{d_j - d_i}{h} \right)^2 \right) \right]^{-\alpha}, \quad (2)$$

with bandwidth $h = 90$ yr and exponent $\alpha = 1$, normalised to mean one. Weighted ridge is implemented as ordinary ridge on rows scaled by $\sqrt{w_i}$.

Variance matching, and what it costs

A regularised regression is a shrinkage estimator: its predictions have systematically smaller variance than the quantity being predicted. Under plain ridge the predicted dates for the anchored corpus span roughly a third of the true range, so no text can be assigned a confidently early or confidently late date — an artifact of shrinkage, not a finding about Hebrew. We correct it by matching the first two moments of the out-of-sample predictions to the true dates,

$$\hat{d} = \bar{d} + S (\hat{d}_{\text{raw}} - \bar{d}_{\text{raw}}), \quad S = \frac{\text{sd}(d)}{\text{sd}(\hat{d}_{\text{raw}})}, \quad (3)$$

with $\bar{d}$, $\text{sd}(d)$, $\bar{d}_{\text{raw}}$ and S all computed from leave-one-book-out predictions on the training corpus only. Here $S = 1.83$.

This is a deliberate trade, and we report both sides of it. Rescaling *increases* mean absolute error: the uncalibrated model achieves 93 yr against 114 yr for the rescaled one, because inflating the spread of predictions amplifies their errors along with their signal. What it buys is a model that is two-sided. Uncalibrated predictions span only 295 yr against 593 yr of real dates, and place just 4 of the 8 genuinely pre-exilic books before the exile; after rescaling the predictions span 539 yr and 6 of 8 are placed correctly.

Since the substantive question in the Pentateuchal debate is which side of the exile a text falls on, a model that can hardly ever say “early” is useless for it whatever its mean absolute error — and its inability to say “early” is a property of the estimator, not evidence that nothing is early. A reader whose interest is point accuracy rather than calibration should use the uncalibrated variant; its figures are in S3 Table.

Conformal prediction intervals

Intervals are conformal, calibrated on the absolute leave-one-book-out residuals of the rescaled predictions. For coverage level $1 - \alpha$ the half-width is the $\lceil (n + 1)(1 - \alpha) \rceil$ -th smallest absolute residual. This is distribution-free and has finite-sample marginal coverage [18, 19]: it assumes only that the target text is exchangeable with the anchored corpus, not that residuals are Gaussian or homoscedastic. On this corpus the half-widths are ± 138 yr at 68% and ± 246 yr at 90%.

For a target unit, $P(\text{post-exilic})$ is computed by adding the empirical residual distribution to the point estimate and taking the fraction of the resulting draws that fall after 586 BCE. This inherits the conformal guarantee and makes no distributional assumption.

Validation

A failure mode that inflates reported accuracy

Reported accuracies for linguistic dating models of this class vary by two orders of magnitude, and part of the variation is a design artifact worth naming, because we committed it ourselves in an earlier version of this work.

In a Bayesian dating model the posterior over a held-out text’s date combines a likelihood from the linguistic features with a prior. If the prior for the held-out text is centred on that text’s own scholarly date — which is the natural thing to do when the prior encodes “what scholars think” — then the model is being asked to recover a number it has already been given. The resulting holdout error measures the width of the prior, not the information in the language.

The effect is large. Under a prior-centred design on this corpus, the median share of posterior precision contributed by the linguistic evidence is 11.7%, and mean holdout error is 44 yr. Under an agnostic prior with an otherwise identical pipeline, the same model achieves 156 yr. For the 8 units whose scholarly dates carry the tightest uncertainty ($\sigma \leq 20$ yr), the linguistic evidence supplies 4.5% of posterior precision and the leaky design reports 6.6 yr against an honest 189 yr (Table 2). A reported accuracy of a decade or two, for a model of this kind, is a signature of the design rather than an achievement.

All results below use an agnostic prior; the reported model is not Bayesian and has no prior to leak.

Leave-one-book-out protocol

Cross-validation folds are whole books, never passages. Passages from one book are strongly correlated, so splitting them across the fold boundary would let the model see the held-out book’s own material. Inside each fold, and using only the training books: features with zero variance or excessive missingness are dropped; medians for imputation are computed; standardisation constants are computed; inverse-density weights are computed; λ is selected by an inner leave-one-book-out loop; and the variance-matching constants and conformal residuals are derived. The held-out book’s date is used only to score the result.

A book’s estimate is the median of its passage estimates. The dispersion of passage estimates within a book is reported separately (Table 3) and is informative in its own right: it is small for compositionally uniform books and large for composites.

What the signal supports

Under this protocol the model achieves a mean absolute error of 114 yr, against 137 yr for the constant predictor that assigns every book the corpus mean date. Rank correlation between true and estimated date is $+0.70$ ($p = 0.0001$), and 74.9% of the 295 book pairs are placed in the correct chronological order.

Because the same corpus was used to choose the passage size and feature set, parametric p -values would be optimistic. Significance is therefore established by a permutation null in which book dates are shuffled and the *entire* pipeline is re-run — weighting, λ selection, variance matching and all — 200 times. Permutation is at book level, not passage level, so the clustering structure is preserved under the null; a passage-level shuffle would break the correlation between passages of the same book and produce a null that

Table 2. Prior leakage inflates reported holdout accuracy. “Prior σ ” is the standard deviation of the scholarly prior in years; “data share” is the fraction of posterior precision contributed by the linguistic features; the two error columns are mean absolute errors in years under a prior centred on the unit’s own scholarly date and under an agnostic prior, with the pipeline otherwise identical.

Unit	Prior σ	Data share	Leaky err.	Honest err.
Haggai	10	2.3%	6	334
Daniel	10	1.7%	3	233
Zechariah_1	10	1.5%	4	236
Amos	20	5.8%	8	144
Habakkuk	20	5.4%	3	51
Isaiah_2	20	8.0%	15	267
Ezekiel	20	5.9%	10	158
Zephaniah	20	5.6%	4	90
Malachi	25	8.9%	18	192
Micah	25	9.7%	3	39
Nahum	25	8.6%	4	48
Hosea	25	9.2%	1	5
Isaiah_1	30	12.0%	16	140
Jer_oracle	30	11.7%	11	105
Chronicles	40	21.8%	20	116
Lamentations	50	35.8%	64	238
Jonah	60	34.1%	18	50
Ezra	60	33.8%	26	80
Esther	60	64.0%	80	200
Nehemiah	60	34.5%	36	116
Obadiah	80	49.2%	29	57
Isaiah_3	100	63.3%	186	296
Ecclesiastes	120	70.7%	150	212
Joel	150	78.3%	200	256
Zechariah_2	150	78.5%	190	242
Median, all	30	11.7%	16	144
Mean, $\sigma \leq 20$ (8 units)	16	4.5%	6.6	189.1

is far too easy to beat. Against this null the observed rank correlation of +0.70 has $p = 0.005$ (null median −0.00), the pairwise ordering accuracy of 74.9% has $p = 0.005$ (null median 51.4%), and the mean absolute error of 114 yr has $p = 0.010$ (null median 179 yr).

We flag one arithmetic trap explicitly, having fallen into it. The 295 pairs formed from 25 books are not independent: each book appears in 25−1 of them. Treating pairwise ordering accuracy as a binomial proportion over 295 independent trials produces p -values wrong by ten orders of magnitude. The permutation figures above are the correct ones.

Two-sidedness and calibration

The property that matters for the Pentateuchal application is whether the model can say “early” as well as “late”. After inverse-density weighting and variance matching, 6 of the 8 genuinely pre-exilic books are placed before 586 BCE and 14 of the 17 genuinely post-exilic books after it, for 20/25 overall (Table 3). Without those two steps the model places essentially nothing before the exile, and its apparent confidence that everything is late is an artifact of the corpus’s date distribution rather than evidence about any text.

Fig 1 shows estimated against true date with the conformal bands. Empirical coverage of the nominal 68% band is slightly conservative, which is the expected behaviour of conformal intervals at this sample size.

What the signal does not support

Three limits deserve statement in the same breath as the results.

First, 114 yr against a 137 yr baseline is a real but modest improvement. The model is a chronological *ordering* device that has been calibrated onto an absolute scale; it is not a precise clock. Reported intervals of ± 138 yr at 68% are the honest expression of that.

Table 3. Leave-one-book-out performance. Dates BCE; error is true minus estimated, so a positive error means the model placed the book too late. “Pass. SD” is the standard deviation of the book’s passage-level estimates, a measure of internal homogeneity. Every quantity is out-of-sample: standardisation, weighting, λ selection and calibration were performed without the held-out book.

Unit	True	Est.	Err.	Pass. SD	$P(\text{post-ex.})$
Amos	760	622	+138	37	0.44
Hosea	745	564	+181	88	0.60
Micah	725	627	+98	70	0.44
Isaiah_1	720	473	+247	92	0.72
Nahum	650	693	-43	0	0.28
Zephaniah	630	660	-30	120	0.32
Habakkuk	605	687	-82	48	0.28
Jer_oracle	605	651	-46	99	0.32
Obadiah	585	617	-32	0	0.52
Ezekiel	580	463	+117	154	0.76
Lamentations	570	707	-137	68	0.24
Isaiah_2	545	669	-124	75	0.28
Haggai	520	553	-33	115	0.60
Zechariah_1	518	432	+86	57	0.84
Malachi	460	517	-57	86	0.64
Joel	400	566	-166	117	0.60
Jonah	400	355	+45	6	0.96
Isaiah_3	400	515	-115	75	0.64
Ezra	380	210	+170	85	1.00
Nehemiah	380	176	+204	83	1.00
Zechariah_2	350	544	-194	51	0.60
Chronicles	330	317	+13	111	1.00
Esther	300	168	+132	53	1.00
Ecclesiastes	250	374	-124	128	0.96
Daniel	167	413	-246	89	0.88

Second, the largest single error is Isaiah 1 at 247 yr. Genre is the main driver: relative to their true dates, narrative books are placed late and poetic books early (Results section). Any single estimate for a text of unusual genre should be read with that in mind.

Third, nothing here resolves dates *within* the post-exilic period. A text estimated at 380 BCE with a ± 138 yr interval is not distinguishable from one estimated at 300 BCE. Claims below are confined to the pre-exilic/post-exilic distinction and to relative ordering, both of which the validation supports.

Results: dating the undated units

The model, fitted on all 25 anchored books, was applied to 19 undated units: the three classical documentary sources, their conventional sub-strata, the five Pentateuchal books as received, the Deuteronomistic prose of Jeremiah, and the three archaic poems. Source partitions follow the standard critical division, specified by chapter and verse in S1 Appendix. No target contributed to training, feature selection, weighting, regularisation or calibration.

The documentary sources are post-exilic

The Priestly source is placed at 330 BCE (68% conformal interval 468–192 BCE) with $P(\text{post-exilic}) = 1.00$; the Deuteronomistic source at 376 BCE (514–238) with 0.96; and the JE composite at 420 BCE (558–282) with 0.84 (Fig 2). All three fall after the exile with probability at least 0.84.

The relative ordering is $\text{JE} \rightarrow \text{D} \rightarrow \text{P}$, the classical sequence of the documentary hypothesis, recovered from linguistic evidence alone by a model trained only on independently anchored prophetic and post-exilic books. We stress that the *ordering* is the better-supported result: given 138 yr intervals, the three sources’ ranges overlap substantially, and the ordering claim rests on the model’s demonstrated ordering ability rather than on the separation of the point estimates.

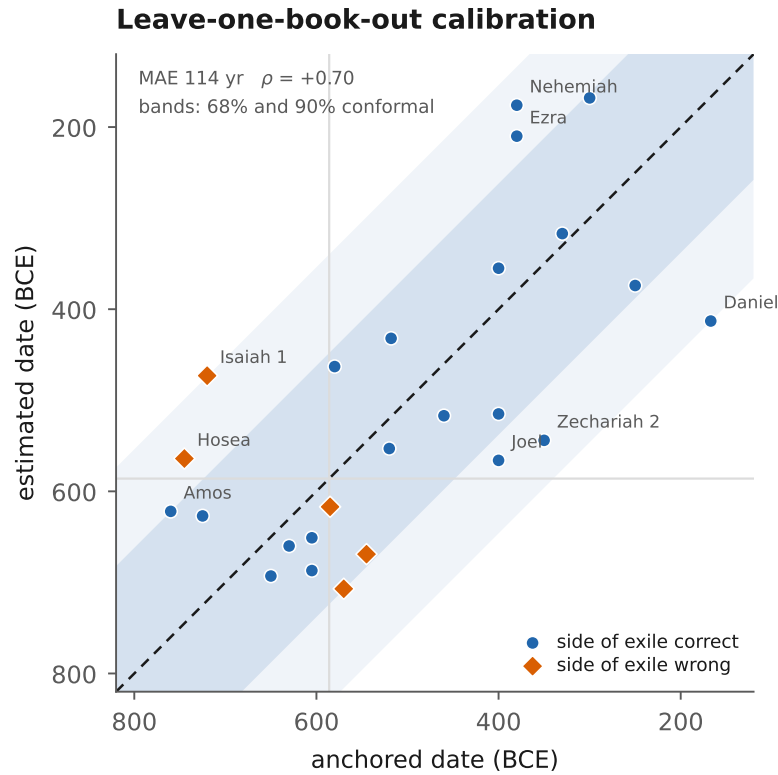


Fig 1. Leave-one-book-out calibration. Each point is one anchored book, dated by a model fitted without it. The dashed line is exact recovery; shaded bands are the 68% and 90% conformal intervals. Books whose estimate falls on the wrong side of the exile are marked with diamonds. Standardisation, weighting, regularisation and interval calibration were all performed inside the fold.

Sub-strata

Splitting Deuteronomy into its law code (chapters 12–26) and its narrative frame places the code at 424 BCE and the frame at 322 BCE — the code earlier than the frame, in the direction the critical consensus expects if the code is the older core around which the frame was later built. Within Leviticus, the Holiness Code (17–26) is placed at 396 BCE against 302 BCE for the Priestly material of 1–16. The JE material of Genesis, Exodus and Numbers is placed at 424, 357 and 442 BCE respectively. The Deuteronomistic prose of Jeremiah is placed at 447 BCE.

These sub-strata differences are small relative to the interval width and should be read as ordering statements consistent with existing literary-critical arguments rather than as independent confirmation of them.

The archaic poems, and an extraction artifact

The three poems traditionally regarded as the oldest Hebrew verse [10, 20, 21] required a correction before they could be dated at all.

Poems in this corpus are conventionally identified by chapter, but the chapters containing them also contain prose. Exodus 15 is 47% prose by word count; Deuteronomy 32 is 26%. Because narrative is systematically placed late by this model, including the prose frame drags the poem’s estimate later. The effect is not small: the Song of the Sea moves from 648 BCE when extracted as a whole chapter to 797 BCE when extracted at verse precision (Exod 15:1–18). We report the verse-precise figures and flag the chapter-level ones as an artifact, since the same convention is widespread.

At verse precision the Song of the Sea is placed at 797 BCE (935–659, $P(\text{post-exilic}) = 0.04$) and the

Table 4. Undated units. Dates BCE; intervals are 68% conformal, calibrated on leave-one-book-out residuals. No unit here contributed to training, feature selection, weighting, regularisation or calibration. The three poems are extracted at verse precision (see text).

Unit	Words	Pass.	Est.	68% interval	<i>P</i> (post-ex.)
<i>Archaic poems</i>					
Song of Deborah	462	1	814	952–676	0.04
Song of the Sea	433	1	648	786–510	0.32
Song of Moses	782	2	529	667–391	0.64
<i>Documentary sources</i>					
JE composite	37,752	74	420	558–282	0.84
<i>Sub-strata</i>					
Genesis JE	22,409	44	424	562–286	0.84
Exodus JE	11,780	23	357	495–219	0.96
Numbers JE	3,563	7	442	579–304	0.84
D source	20,128	39	376	514–238	0.96
D law code	7,353	14	424	562–286	0.84
D frame	11,993	23	322	460–184	1.00
P source	54,435	106	330	468–192	1.00
Leviticus P	10,496	21	302	440–164	1.00
Holiness Code	5,958	12	396	534–258	0.92
Jeremiah Dtr prose	15,197	30	447	584–309	0.84
<i>Pentateuch as received</i>					
Genesis	28,764	56	413	551–275	0.88
Exodus	23,748	47	347	485–209	0.96
Leviticus	17,099	33	328	466–190	1.00
Numbers	23,188	45	352	490–214	0.96
Deuteronomy	20,128	39	376	514–238	0.96

Song of Deborah at 823 BCE (961–685, 0.04). These are the two earliest of the 19 undated units, in agreement with a century of philological argument that they are the oldest surviving Hebrew, and reached without training on them.

The Song of the Sea’s prose continuation (Exod 15:19–27) is placed at 470 BCE, 327 yr later than the poem it follows. A gap of that size between adjacent verses is itself evidence that the poem is not a composition of its narrative frame. The Archaizing section takes up whether that gap could have been manufactured.

The Song of Moses (Deut 32:1–43) is placed at 610 BCE (747–472, 0.56), appreciably later than the other two and straddling the exile. This is not what a reading of its content would predict: its divine-council theology, in which Elyon apportions the nations among the gods and YHWH receives Israel as his share (32:8–9), is usually taken as among the most archaic material in the Hebrew Bible. The model sees its language, not its theology, and places the language late. We report the discrepancy rather than explain it away.

Genre is the dominant confound

Residuals on the anchored corpus are structured by genre. Relative to their true dates, narrative books are placed *late* and poetic books *early*. The clearest case is Lamentations, whose date is fixed by its subject matter — it laments the destruction of 586 BCE — and which is therefore as securely anchored as any text in the corpus. The model places it at 707 BCE against a true 570 BCE, an error of 137 yr in the direction of greater antiquity. Lamentations is poetry.

This has a direct bearing on the poems above. If poetry as such is placed early by this model, then part of the antiquity assigned to the Songs of the Sea and Deborah may be genre rather than date. We decline to apply a genre correction: with a single securely dated poetic book in the anchored corpus, any correction would be estimated from one observation. The appropriate statement is that the poems’ early placement is supported by the model but partially confounded with genre, and that enlarging the anchored poetic corpus is the way to separate them.

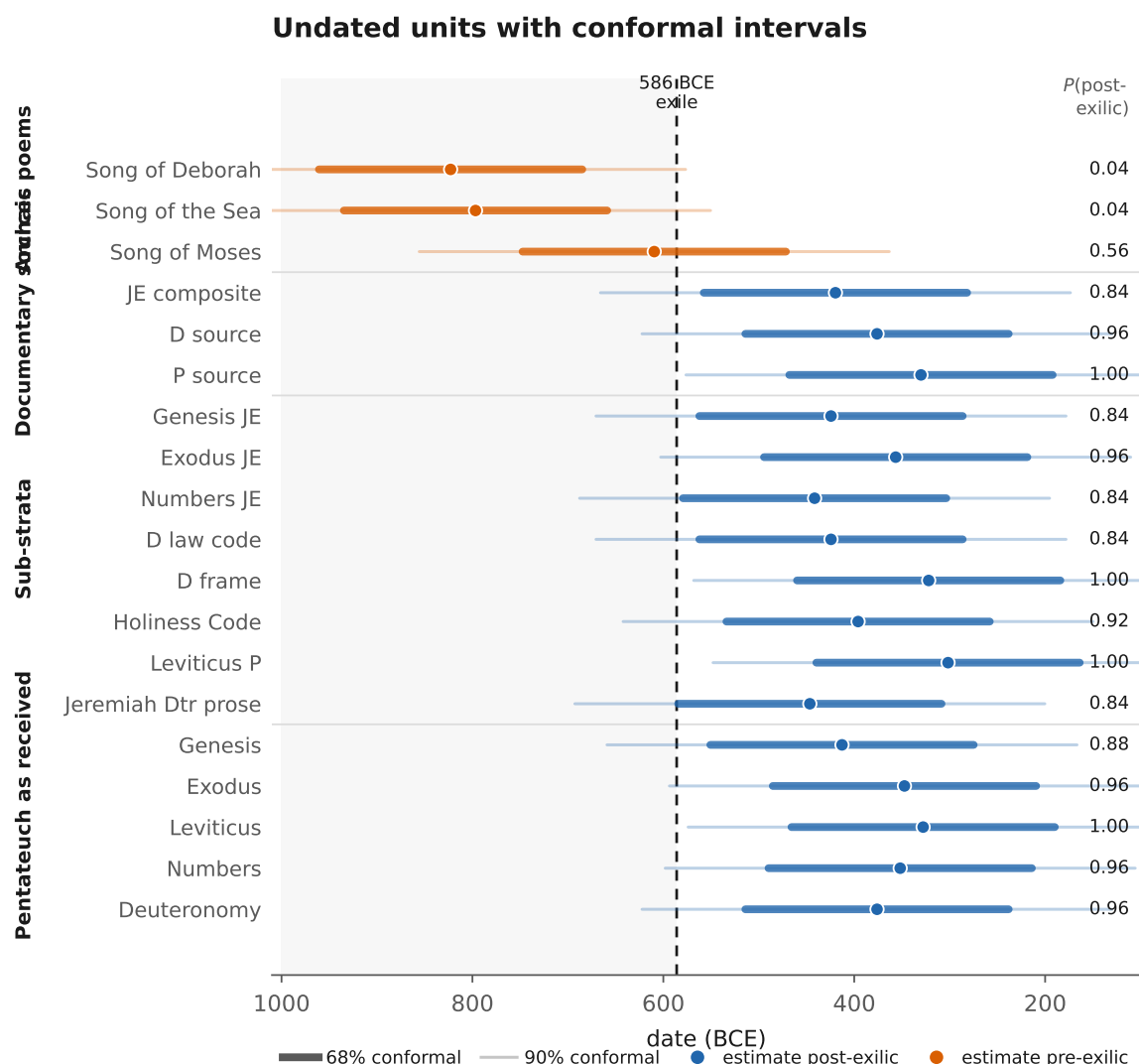


Fig 2. Undated units with conformal intervals. Thick bars are 68% intervals, thin bars 90%; the dashed line is the exile of 586 BCE. Colour marks which side of the exile the point estimate falls on. The three poems are extracted at verse precision. No unit shown contributed to training, feature selection, weighting, regularisation or calibration.

How much antiquity can archaizing buy?

Everything above depends on a premise: that the language of a text reflects when it was written rather than when its author wished to seem to be writing. This section measures the size of the gap between those two things, first by performing archaizing ourselves on Hebrew, then by using Greek texts whose authors performed it for us.

Synthetic archaizing in Hebrew

Take a securely dated *late* book, replace its LBH forms with their CBH counterparts at a controlled rate r , re-extract all 579 features from the modified text, and re-date it with a model trained on the other 24 books. Because we performed the substitution, the ground truth is known exactly.

The substitutions are the classical diagnostics of the CBH/LBH literature, applied in the LBH $\rightarrow$ CBH direction: the relative particle *še-* $\rightarrow$ *ašer*; the first-person pronoun *anī* $\rightarrow$ *ankī*; *malkūt* $\rightarrow$ *mamlākāh*; and

the negator ' $\bar{e}n \rightarrow \bar{l}\bar{o}$ ', which changes part of speech as well as lexeme. Substitution is applied at the token level before feature extraction, so every downstream feature — lexeme rates, part-of-speech transitions, phrase types, clause relations — sees the altered text exactly as it would if a scribe had written it that way. r runs from 0 to 1; at $r = 1$ not one LBH token of these types survives anywhere in the book.

Table 5. What total lexical archaizing buys. Each securely dated late book had its Late Biblical Hebrew forms replaced by Classical counterparts at rate r and was then re-dated by a model trained on the other 24 books. $r = 1$ replaces every eligible token in the book. Dates BCE; a positive shift means the book appears older. Ezra, Nehemiah and Chronicles offer too few eligible tokens for their null results to be informative.

Book	True	$r = 0$	$r = 1$	Shift	Swaps	per 1k w
Chronicles	330	275	275	+0	94	2.7
Esther	300	126	135	+9	42	9.1
Ecclesiastes	250	386	417	+31	142	33.5
Daniel	167	384	409	+25	47	13.7
Ezra	380	179	182	+3	13	2.5
Nehemiah	380	139	139	-0	28	3.6
Mean apparent shift				+11		

Total lexical archaizing moves the estimated date by a mean of +11 yr, with a maximum of +31 yr across the 6 books (Table 5). The response is gradual rather than thresholded: there is no substitution rate at which the estimate jumps.

Three of the six books had almost nothing to substitute — Ezra offers 13 eligible tokens in the entire book — so their null results carry little information; one cannot archaize a text that already lacks the forms. Among the 3 books with substantial substitution density the exchange rate is about 1.3 yr of apparent antiquity per substitution per 1000 words. Ecclesiastes at $r = 1$ has 33.5 substitutions per 1000 words — roughly one word in thirty replaced, a density no scribe sustains across a book — and moves +31 yr.

Where the model's leverage lives

The reason for that small number is visible in the model's coefficients.

Fig 4 reports the change in estimated date produced by a one-standard-deviation shift in each feature. Lexical features carry 47% of total leverage — but spread across 250 features, none worth more than 12.2 yr per standard deviation. No single word is a lever. The remaining leverage sits in part-of-speech transition rates, verbal aspect and stem distributions, and clause-relation profiles: high-frequency grammatical habits rather than vocabulary choices.

These are precisely the features that the passage-level design forces the model to rely on. Requiring a feature to be measurable within 500 words demotes the rare diagnostic vocabulary — which is the manipulable part of the language — and promotes high-frequency morphology, which is not. Resistance to archaizing was not designed into this model; it is a consequence of the sample-size constraint.

The quantitative form of the statement: because the model is linear in standardised features, the smallest coordinated feature shift that displaces an estimate by D years has length $D/\|\beta\|$ in standard deviations. For the 327 yr gap between the Song of the Sea and its prose frame, that is 5.2 standard deviations across 579 dimensions. Not a word swap — a systematic re-profiling of the text's morphology and syntax, sustained across the whole passage. The Greek results below show that this is achievable, by authors trained to achieve it.

Greek: archaizing that really happened

The Hebrew experiment simulates archaizing, and simulates only its lexical component. Ancient Greek supplies the real thing, labelled. During the Second Sophistic, prose authors of the first three centuries CE deliberately wrote in imitation of Classical Attic, four to seven centuries after the fact. The programme was explicit and institutional: Attic prose composition was taught in the rhetorical schools, dictionaries of permitted Attic usage were compiled, and the imitation was successful enough that its practitioners were

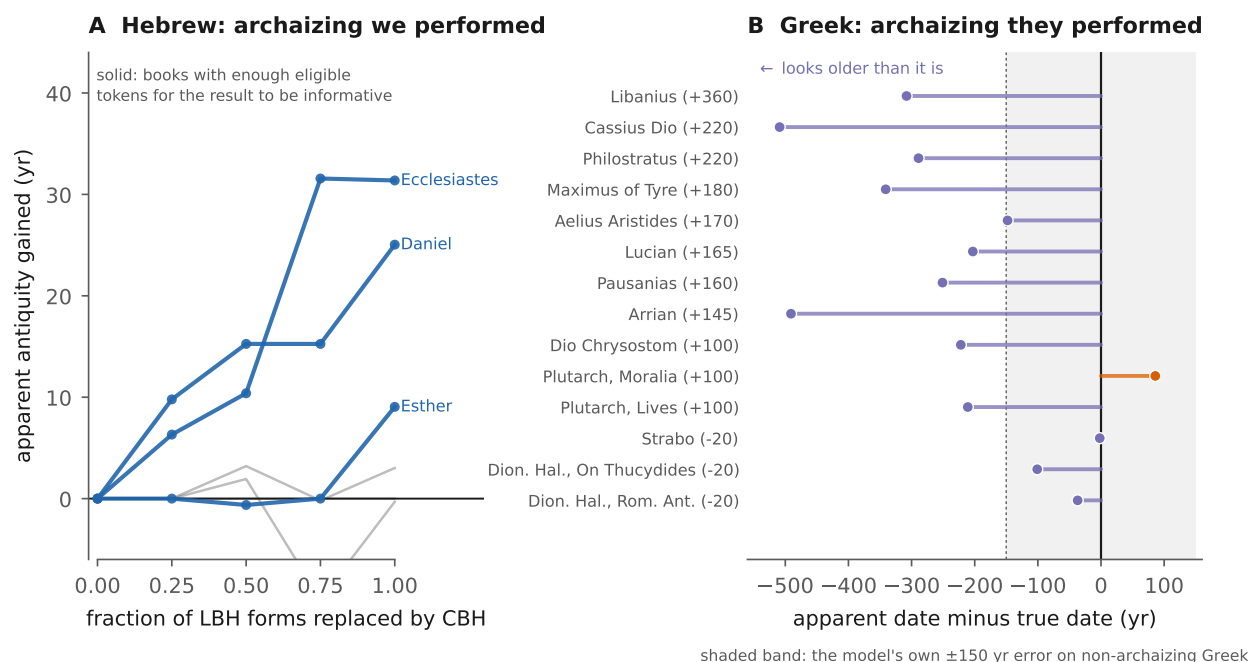


Fig 3. The price of archaizing, by mechanism. **A**, Hebrew: apparent antiquity gained as Late Biblical Hebrew forms are replaced by Classical counterparts at increasing rates in six securely dated late books. Solid lines are the books with enough eligible tokens for the result to be informative. **B**, Greek: displacement of the Second Sophistic Atticizers under a model trained only on non-archaizing Greek, with the model's own error on non-archaizing texts shaded for comparison. Lexical substitution moves an estimate by tens of years; reconstructed grammar moves it by hundreds.

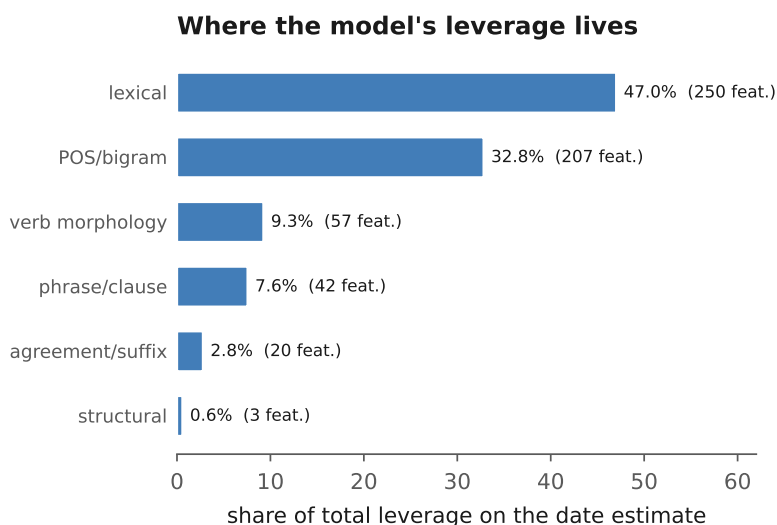


Fig 4. Where the model's leverage lives. Share of total leverage on the date estimate by feature family, where leverage is the absolute change in estimated date produced by a one-standard-deviation shift in a feature, summed within family. Lexical features carry the largest share but spread it across 250 weak features; the majority of leverage sits in morphology and syntax.

read as models of style for a millennium. Their composition dates are securely known from external evidence. This is the labelled archaizing set that Hebrew cannot provide.

We built the same pipeline on a 42-text corpus of non-archaizing Greek prose spanning the Classical and

Koine periods, partitioned into 4263 passages, with 747 features. Greek lacks a morphological annotation comparable to BHSA, so hand-built morphosyntactic rates (optative, infinitive and participle rates, particle and negation rates, the $\tau\tau$ versus $\sigma\sigma$ contrast, sentence length) are supplemented by word-form, character n-gram and function-word bigram rates — all high-frequency by construction, as on the Hebrew side. Under leave-one-text-out the Greek model achieves 150 yr against 238 yr for the constant predictor, rank correlation +0.65, and 71.7% pairwise ordering accuracy. The replication is therefore genuine: the same design recovers a diachronic signal in an unrelated language.

The 14 Atticizing texts were excluded from training entirely and then dated.

Table 6. Second Sophistic Atticizers under a model trained only on non-archaizing Greek. Dates CE (negative = BCE). A negative shift means the text is dated earlier than it was written, i.e. the archaizing worked. These 14 texts were excluded from training entirely.

Author	Work	True	Est.	Shift	Pass.
Dionysius of Halicarnassus	Roman Antiquities	-20	-57	-37	500
Dionysius of Halicarnassus	On Thucydides	-20	-121	-101	4
Strabo	Geography	-20	-22	-2	268
Plutarch	Parallel Lives (selected)	+100	-111	-211	4
Plutarch	Moralia (selected)	+100	+186	+86	26
Dio Chrysostom	Orations (selected)	+100	-122	-222	308
Arrian	Anabasis of Alexander	+145	-346	-491	140
Pausanias	Description of Greece	+160	-91	-251	223
Lucian	True History and selected dialogues	+165	-38	-203	5
Aelius Aristides	Sacred Tales	+170	+22	-148	4
Maximus of Tyre	Orations (Dialexeis)	+180	-161	-341	8
Philostratus	Lives of the Sophists	+220	-69	-289	42
Cassius Dio	Roman History (epitome books)	+220	-289	-509	262
Libanius	Orations (selected)	+360	+52	-308	216
Mean displacement				-216	

The mean displacement is -216 yr, with 13 of 14 texts placed earlier than they were written (Fig 3B). Cassius Dio is displaced -509 yr. Set against the model’s own 150 yr error on non-archaizing texts, this is a large effect: deliberate, professionally trained archaizing displaces a text by more than one full error width, and in the worst cases by half a millennium.

Two features of the Greek corpus deserve comment because they bear on how the number should be read.

First, the effect is if anything *understated*, because the training set is contaminated. The corpus manifest classifies Aelian, Appian, Diogenes Laertius, Plotinus, Porphyry, Iamblichus and the sophistic novelists as Koine, so they were used as non-archaizing training material — yet every one of them is described in the literary-historical record as classicizing, and the model placed them among the largest early displacements in the training set even though it had been told they were not archaizing. A training set that contains archaizers teaches the model that late texts look early, which shrinks the measured displacement of the declared Atticizers. Because the classification is a scholarly judgment rather than a datum, we do not adopt one and defend it; we report the measurement under three definitions of the training set — the manifest’s own labels, a stricter classification that reclassifies the authors named above, and a date-blind variant that discards register labels entirely and trains only on texts composed before 50 CE, prior to the Second Sophistic (S6 Table).

The result is robust in sign and grows under stricter definitions: the mean displacement is -216 yr under the manifest’s labels, -306 yr under the stricter classification, and -412 yr under the date-blind variant, with 45 of 47 individual texts across the three placed earlier than they were written. Two things are worth noting about the stricter variant. It removes eight texts from training, and yet the model’s own leave-one-text-out error *falls* from 150 yr to 114 yr and its rank correlation rises from +0.65 to +0.76 — removing suspected archaizers from the training set makes the date model better, which is what one would expect if they were genuinely mislabelled. We nonetheless report the manifest’s own labels as the primary specification throughout, because it is the only one that involves no judgment of ours, and it is the most conservative.

Second, the two Atticizers displaced least — Strabo and Dionysius of Halicarnassus — are the earliest in the set, writing around the turn of the era when the Atticizing programme was young. The displacement

grows with the maturity of the movement, which is what one would expect if it is measuring imitation rather than noise.

The available texts vary greatly in length, from four passages to five hundred, and a four-passage estimate is correspondingly noisy. The result is not driven by the short texts: weighting each text by its passage count gives -213 yr, and restricting to the 8 texts with at least 40 passages gives -264 yr, with all 8 placed earlier than they were written.

What the two experiments jointly establish

The two experiments do not give the same number, and the difference is the substantive finding.

Total *lexical* archaizing in Hebrew buys +11 yr. Real *morphosyntactic* archaizing in Greek buys -216 yr. The gap is roughly an order of magnitude, and it falls exactly where the feature-leverage analysis said it would. Swapping diagnostic vocabulary moves a handful of the 250 weak lexical features and leaves the other half of the model's leverage untouched. The Atticizers did not swap vocabulary; they reconstructed a grammar — reviving the optative, restoring the $\mu\epsilon\nu \dots \delta\epsilon$ articulation, writing $\tau\tau$ for $\sigma\sigma$, matching Classical participial density — which is precisely the coordinated multi-dimensional shift that the Hebrew sensitivity analysis identified as the only route to a displacement of centuries. They achieved it because an entire educational apparatus existed to teach them how.

The conclusion is therefore conditional rather than absolute, and we state it in that form. Archaizing can defeat a model of this class. What it takes is not a scribe who knows the old words but a scribe trained to write the old grammar, sustained across a whole composition. The relevant question for any particular text is whether such training existed in its milieu.

For the Hebrew Bible this becomes a historical question rather than a statistical one, and it is not one this paper settles. We note only that the institutional preconditions differ sharply between the two cases: the Second Sophistic had a documented curriculum of Attic prose composition, lexica of permitted usage, and a literary culture that rewarded successful imitation, while no comparable programme of Classical Hebrew prose composition is attested for the Second Temple period. The Qumran evidence adduced by the dialectal school [5] is the closest analogue and is a good deal thinner: the sectarian scrolls mix CBH forms with Aramaisms rather than sustaining a consistent Classical grammar, which on the evidence here would displace their apparent date by tens of years, not hundreds.

Applied to the results of the previous section, this means the early placement of the Song of the Sea and the Song of Deborah is not immune to an archaizing explanation, and we withdraw any such claim. It means that an archaizing explanation must posit sustained morphosyntactic imitation of a kind for which Second Temple Judaism offers no attested institutional support, rather than the lexical substitution that the CBH/LBH literature actually describes. That is a weaker defence than immunity, and it is the one the evidence supports.

Discussion

Limitations

Interval width. The 68% intervals are ± 138 yr and the 90% intervals ± 246 yr. These are honest but wide, and they preclude any claim about position within the Persian period. Every substantive claim above is either a side-of-the-exile claim or an ordering claim.

Genre. Poetic texts are placed early and narrative texts late relative to their true dates. With one securely dated poetic book in the anchored corpus, genre cannot be separated from date by this design. This directly affects the archaic poems, which are the results most likely to be cited and are also the ones most exposed to this confound.

Corpus asymmetry. The anchored corpus is skewed post-exilic, because that is where external synchronisms are dense. Inverse-density weighting mitigates but does not remove the skew, and the pre-exilic end of the scale rests on 8 books. Extending the anchored corpus at the early end would do more for this model than any change of estimator.

The two archaizing measurements bound different things. Four substitution pairs is a small vocabulary of archaizing, so the Hebrew experiment prices only the lexical mechanism, and prices it at the extreme setting of total replacement. The Greek experiment prices real imitation but only at the intensity Second Sophistic authors actually achieved, in a language whose annotation is coarser than BHSA, and against a training set whose register classification is a scholarly judgment rather than a datum (S6 Table reports the sensitivity to that judgment). Neither experiment prices sustained morphosyntactic archaizing *in Hebrew*, which is the quantity a critic of the Song of the Sea result would most want. Simulating it would require a model of what Classical Hebrew grammar an imitator could reconstruct, which we do not have.

The Greek transfer is an analogy, not a measurement of Hebrew. That Greek imitators achieved a -216 yr displacement establishes that the deep mechanism works in principle and gives its rough scale. It does not establish that a Hebrew scribe could achieve the same, and the argument that Second Temple Judaism lacked the institutional apparatus is a historical claim we assert rather than demonstrate.

One database, one text tradition. All Hebrew results depend on the BHSA annotation of the Masoretic text. Consonantal variants in the Qumran and Septuagint traditions are not represented, and the Masoretic vocalisation reflects a much later stage of the language than any composition date discussed here. The features used are drawn from the consonantal text and its morphological parse, but the parse itself is a modern scholarly product.

Scholarly context for late Torah dating

The finding that the Pentateuchal sources are post-exilic is not novel in biblical scholarship, though it remains contested. A substantial tradition places the final composition of P in the Persian period [22, 23], and the “Persian period Pentateuch” position associated with the Copenhagen and Sheffield schools places the whole compositional process there [24, 25]. Against this, Hurvitz’s linguistic argument for a pre-exilic P [1, 9] and the broader neo-documentary position maintain earlier dates.

The contribution here is not the direction of the result but that it arrives with a calibrated uncertainty and a measured archaizing sensitivity. The linguistic argument for a pre-exilic P has always had to assume that the post-exilic appearance of a text cannot be manufactured; the argument for a post-exilic P has always had to assume the converse. The measurements above constrain both.

Extensibility

The pipeline is language-agnostic given a morphologically annotated corpus and a set of externally anchored dates. The Greek replication was implemented from the same code with a different feature extractor. The binding constraint is annotation, not method: Ugaritic, Phoenician and epigraphic Hebrew would supply badly needed early anchors but lack comparable morphological parsing. Producing such a parse for the Iron Age epigraphic corpus would be the single most valuable input to this line of work.

Relationship to authorship attribution

Attribution and dating are complementary. A method that assigns a text to a scribal school and a method that assigns it a date constrain different axes of the same question, and their errors are not the same errors. A natural extension is to use attribution output as a register covariate in the date model, which would allow the genre confound documented above to be modelled rather than merely reported.

Conclusion

Biblical Hebrew morphosyntax carries a diachronic signal strong enough to order books in time and to distinguish pre-exilic from post-exilic composition, and too weak to resolve dates within the post-exilic period. Applied to the Pentateuchal sources, a model calibrated on independently anchored books places the Priestly, Deuteronomic and JE composites after the exile with probability at least 0.84, and recovers the classical relative sequence $JE \rightarrow D \rightarrow P$ without having been trained on any of them. The Song of the Sea and the Song of Deborah are the earliest of the 19 undated units.

The premise on which all such conclusions rest — that a late author cannot convincingly write early — has been treated as an assumption for two decades. It is measurable, and measuring it shows that it is not one assumption but two, with different answers. Complete lexical archaizing of a securely dated late Hebrew book buys a mean of +11 yr of apparent antiquity: the diagnostic vocabulary that the CBH/LBH literature treats as the substance of archaizing turns out to carry very little of a passage-level model’s leverage. The Second Sophistic Atticizers, who reconstructed Classical grammar rather than Classical vocabulary and were trained to do so by a formal curriculum, are displaced by -216 yr.

Archaizing therefore has a price, and the price is set by how deep the imitation goes. This is a more useful result than immunity would have been, because it converts a rhetorical impasse into a comparison that can be made case by case. For any text whose date is disputed on linguistic grounds, the question is no longer whether archaizing is possible in the abstract but whether the compositional milieu in question supported the deep kind. For the Second Sophistic the answer is documented and affirmative. For Second Temple Judaism no comparable programme is attested, and the closest analogue — the Qumran sectarian scrolls, which mix Classical forms with Aramaisms rather than sustaining a Classical grammar — resembles the cheap mechanism rather than the expensive one.

Supporting information

Table 7. S1 Appendix. Source partitions. Chapter specification of every undated unit. The five Pentateuchal books as received are omitted (they are simply the whole book). Partitions follow the standard critical division; the three poems were additionally re-extracted at verse precision.

Unit	Chapters
P source	Genesis 1, 5, 6, 9, 11, 17, 23, 25, 35, 36, 46, 48; Exodus 6–7, 12, 16, 25–31, 35–40; Leviticus 1–27; Numbers 1–10, 13–20, 25–36
JE composite	Genesis 2–4, 6–8, 12–16, 18–22, 24–34, 37–45, 49–50; Exodus 1–5, 8–11, 13–14, 17–24, 32–34; Numbers 11–12, 21–24
D source	Deuteronomy 1–34
Genesis JE	Genesis 2–4, 6–8, 12–16, 18–22, 24–34, 37–45, 49–50
Exodus JE	Exodus 1–5, 8–11, 13–14, 17–24, 32–34
Numbers JE	Numbers 11–12, 21–24
D law code	Deuteronomy 12–26
D frame	Deuteronomy 1–11, 27–31, 33–34
Song of Moses	Deuteronomy 32 <i>Deut 32:1–43 at verse precision</i>
Holiness Code	Leviticus 17–26
Leviticus P	Leviticus 1–16
Song of the Sea	Exodus 15 <i>Exod 15:1–18 at verse precision (see text)</i>
Song of Deborah	Judges 5 <i>Judg 5:2–31 at verse precision</i>
Jeremiah Dtr prose	Jeremiah 7, 11, 17–18, 21, 24–29, 32–45, 52

Table 8. S2 Table. Effective sample size by passage size. Passages within a book are correlated, so the raw passage count overstates the information available. $n_{\text{eff}} = n / (1 + (\bar{n} - 1)\rho_I)$ with ρ_I the median intraclass correlation across features.

Passage size	Passages	Median ICC	Design effect	n_{eff}	Gain vs $n = 25$
300	470	0.09	2.68	175.3	7.01x
500	285	0.13	2.38	119.6	4.79x
1000	145	0.20	1.96	74.0	2.96x

S1 Appendix. Source partitions. Chapter-and-verse specification of every documentary source, sub-stratum and poem dated in this study (Table 7).

Table 9. S3 Table. Configuration search. Out-of-sample metrics for each combination of passage size, feature set and calibration under leave-one-book-out. “Coverage” is the empirical coverage of the nominal 68% interval; values near 1.0 indicate conservative intervals and values near 0.5 indicate the shrinkage problem discussed in the Model section.

Passage	Features	<i>p</i>	Calib.	MAE	ρ	Pair %	Coverage
1000	base	579	False	105	+0.56	69.5	0.60
1000	base	579	True	97	+0.56	68.5	0.64
1000	both	1268	False	94	+0.62	71.5	0.58
1000	both	1268	True	96	+0.62	71.9	0.68
1000	ngram	689	False	96	+0.59	70.8	0.53
1000	ngram	689	True	99	+0.60	70.8	0.67

Table 10. S6 Table. Register sensitivity of the Greek archaizing measurement. The displacement of the Atticizers depends on which texts are treated as non-archaizing training material. Variant A uses the corpus manifest’s own labels; B additionally reclassifies authors independently described as classicizing in the literary-historical record; C discards register labels entirely and trains only on texts composed before 50 CE, prior to the Second Sophistic. Dates CE; a negative shift means the text is dated earlier than it was written.

Training set	Texts	Tested	MAE	ρ	Mean shift	Too early
A as-received manifest labels	42	14	150	+0.65	-216	13/14
B strict: classicizing authors reclassified	34	22	114	+0.76	-306	22/22
C date-blind: train only on j= 50 CE	26	11	121	+0.62	-412	10/11

S2 Table. Power analysis. Intraclass correlation, design effect and effective sample size by passage size (Table 8). 565 566

S3 Table. Configuration search. Combinations of passage size, feature set and calibration with out-of-sample metrics, including the uncalibrated variants (Table 9). 567 568

S4 Table. Feature leverage. Years of apparent date per standard-deviation shift for every feature, by family. 569 570

S5 Table. Greek corpus. All texts with dates, registers, genres and sources. 571

S6 Table. Register sensitivity of the Greek archaizing measurement. Displacement of the Atticizers under three definitions of the non-archaizing training set (Table 10). 572 573

Data availability 574

All code, extracted feature matrices, and result files are available at the repository cited in the 575
acknowledgments. The BHSA database is available under its own licence from the Eep Talstra Centre for 576
Bible and Computer. Greek texts derive from the open First1KGreek and Perseus collections. 577

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References

1. Hurvitz A. A Linguistic Study of the Relationship between the Priestly Source and the Book of Ezekiel: A New Approach to an Old Problem. No. 20 in *Cahiers de la Revue Biblique*. Paris: Gabalda; 1982.
2. Rooker MF. Biblical Hebrew in Transition: The Language of the Book of Ezekiel. No. 90 in *Journal for the Study of the Old Testament Supplement Series*. Sheffield: JSOT Press; 1990.
3. Polzin R. Late Biblical Hebrew: Toward an Historical Typology of Biblical Hebrew Prose. No. 12 in *Harvard Semitic Monographs*. Missoula: Scholars Press; 1976.
4. Horrocks G. *Greek: A History of the Language and Its Speakers*. 2nd ed. Oxford: Wiley-Blackwell; 2010.
5. Young I, Rezetko R, Ehrensverd M. *Linguistic Dating of Biblical Texts*. vol. 2. London: Equinox; 2008.
6. Rezetko R, Young I. Historical Linguistics and Biblical Hebrew: Steps toward an Integrated Approach. No. 9 in *Ancient Near East Monographs*. Atlanta: SBL Press; 2014.
7. Hurvitz A. The Historical Quest for ‘Ancient Israel’ and the Linguistic Evidence of the Hebrew Bible: Some Methodological Observations. *Vetus Testamentum*. 1997;47:301-15.
8. Young I, editor. *Biblical Hebrew: Studies in Chronology and Typology*. No. 369 in *Journal for the Study of the Old Testament Supplement Series*. London: T&T Clark; 2003.
9. Hurvitz A. A Concise Lexicon of Late Biblical Hebrew: Linguistic Innovations in the Writings of the Second Temple Period. No. 160 in *Supplements to Vetus Testamentum*. Leiden: Brill; 2014.
10. Robertson DA. Linguistic Evidence in Dating Early Hebrew Poetry. No. 3 in *SBL Dissertation Series*. Missoula: Scholars Press; 1972.
11. Schniedewind WM. *How the Bible Became a Book: The Textualization of Ancient Israel*. Cambridge: Cambridge University Press; 2004.
12. Mosteller F, Wallace DL. *Applied Bayesian and Classical Inference: The Case of The Federalist Papers*. 2nd ed. New York: Springer; 1984.
13. Holmes DI. The Evolution of Stylometry in Humanities Scholarship. *Literary and Linguistic Computing*. 1998;13:111-7.
14. Burrows J. ‘Delta’: A Measure of Stylistic Difference and a Guide to Likely Authorship. *Literary and Linguistic Computing*. 2002;17:267-87.
15. Faigenbaum-Golovin S, Kipnis A, Bühler A, Piasetzky E, Römer T, Finkelstein I. Critical Biblical Studies via Word Frequency Analysis: Unveiling Text Authorship. *PLOS One*. 2025;20(6):e0322905. doi:10.1371/journal.pone.0322905.
16. Eep Talstra Centre for Bible and Computer. *Biblia Hebraica Stuttgartensia Amstelodamensis (BHSA)*, version 2021; 2021. Amsterdam: ETCBC, Vrije Universiteit Amsterdam. <https://github.com/ETCBC/bhsa>.
17. Roorda D. Coding the Hebrew Bible. *Research Data Journal for the Humanities and Social Sciences*. 2018;3:27-41.

18. Vovk V, Gammerman A, Shafer G. Algorithmic Learning in a Random World. New York: Springer; 2005. 621
622
19. Lei J, G'Sell M, Rinaldo A, Tibshirani RJ, Wasserman L. Distribution-Free Predictive Inference for Regression. *Journal of the American Statistical Association*. 2018;113(523):1094-111. 623
624
20. Albright WF. Yahweh and the Gods of Canaan: A Historical Analysis of Two Contrasting Faiths. London: Athlone Press; 1968. 625
626
21. Cross FM, Freedman DN. Studies in Ancient Yahwistic Poetry. vol. 21 of Society of Biblical Literature Dissertation Series. Missoula, MT: Scholars Press; 1975. 627
628
22. Blenkinsopp J. An Assessment of the Alleged Pre-Exilic Date of the Priestly Material in the Pentateuch. *Zeitschrift für die alttestamentliche Wissenschaft*. 1996;108(4):495-518. 629
630
23. Nihan C. From Priestly Torah to Pentateuch: A Study in the Composition of the Book of Leviticus. vol. 25 of *Forschungen zum Alten Testament 2. Reihe*. Tübingen: Mohr Siebeck; 2007. 631
632
24. Davies PR. In Search of 'Ancient Israel'. No. 148 in *Journal for the Study of the Old Testament Supplement Series*. Sheffield: JSOT Press; 1992. 633
634
25. Lemche NP. The Old Testament — A Hellenistic Book? *Scandinavian Journal of the Old Testament*. 1993;7(2):163-93. 635
636